

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester III Examination 2025-26 (NEP)

DEPARTMENT OF GEOLOGY**Value Added Course (VAC)****Paper Code: ESMV31 — Medical Geology**

Time: 2:30 Hours

Maximum Marks: 70

Ref Code: CECNF/N/2025-26/050

*Instruction: Write your Roll No. at the top immediately on the receipt of this question paper.**Note: Answer five questions in all, selecting at least two questions each from Section–A and Section–B. Section–C is compulsory. Figures in the right-hand margin indicate marks.***SECTION – A****Question 1**

[14 Marks]

What is a mineral nutrient? Explain in detail why mineral elements are needed for good human health.

Question 2

[14 Marks]

Define the hydrological cycle and discuss in detail the various components of the hydrological cycle with a neat sketch.

Question 3

[14 Marks]

Discuss the biological response of elements with appropriate examples.

SECTION – B**Question 4**

[14 Marks]

Write short notes on any **two** of the following ($2 \times 7 = 14$ Marks):

- (a) Environmental epidemiology with reference to geogenic factors and disease mapping
- (b) Toxicology risk management
- (c) Geophagy in humans, its major causes, potential benefits, and associated health risks

Question 5

[14 Marks]

Write short notes on any **two** of the following ($2 \times 7 = 14$ Marks):

- (a) Uptake and regulation of iron in the human body
- (b) Drawbacks and health impacts of arsenic consumption on human health
- (c) Application of geochemistry to environmental health issues

Question 6

[14 Marks]

Write short notes on any **two** of the following ($2 \times 7 = 14$ Marks):

- (a) Biological functions of major elements
- (b) How fluoride overdose in drinking water harms human health
- (c) Diversity in natural distribution and abundance of elements

SECTION – C (Compulsory)**Question 7**

[14 Marks]

Answer all the following questions:

- (a) Write the names of two minerals present in human bone. [2 Marks]
- (b) What are the four major elements of the human body? [2 Marks]
- (c) What is nitrogen fixation? [1 Mark]
- (d) Define Total Dissolved Solids (TDS) and Electrical Conductivity (EC) parameters of water. [2 Marks]
- (e) What is the permissible limit of nitrate for drinking water? [1 Mark]
- (f) What are the carcinogenic associations of fibrous minerals (e.g., asbestos) in the human body? [3 Marks]
- (g) Write the geochemical associations of elements in major igneous and sedimentary rock types. [3 Marks]